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Ahmad Dhuha Habibullah

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ABOUT ME

Earth scientist and physical oceanographer with a Master's degree in Earth Science and four years of experience as a system integration specialist for marine-meteorological infrastructure and operational forecasting. Specialist in marine heatwaves (MHWs), climate variability and large-scale gridded ocean datasets, processed in Python and Matlab.

EDUCATION AND TRAINING

01/08/2020 – 31/07/2022 Bandung, Indonesia

MASTER OF SCIENCE IN EARTH SCIENCE (OCEANOGRAPHY) Bandung Institute of Technology

Thesis: Analyzing Marine Heatwaves in Indonesia (1982–2020). Processed 38 years of daily high-resolution NOAA OISST data in Python to characterise MHW duration, frequency and intensity across Indonesian waters.

Level in EQF EQF level 7

01/08/2015 – 31/08/2019 Bengkulu, Indonesia

BACHELOR OF SCIENCE IN PHYSICS Universitas Bengkulu

Thesis: Development of a Microcontroller-Based Solar Irradiance Monitoring Instrument using Photovoltaic Panel. Electronics, programming, and design, testing and calibration of metocean instruments (surface current drifter, tide gauge, pyranometer

Level in EQF EQF level 6

WORK EXPERIENCE

SYSTEM INTEGRATION SPECIALIST – PT SIBACIPTA TELEKOMINDO – 01/09/2022 – Current – INDONESIA

Marine weather station and vessel network (40 coastal stations, 2022–2023)

- Field technician and deployment coordinator for 40 automated coastal and port marine weather stations across Indonesia, including remote islands (Larantuka, Rote, Saumlaki, Manokwari, Tual, Merauke, Tarakan, Pontianak, Ulee Lheue and others): site surveys, civil structure assessments, hardware assembly, sensor calibration and handover.
- Designed custom hardware brackets for optimal sensor placement in harsh marine environments.
- Led installation teams and coordinated supplier partners and clients; kept multi-site deployments to schedule and technical specification.
- Installed Vessel Automatic Weather Stations (VAWS) on commercial passenger and cargo vessels; contributed to equipment integration and long-term maintenance planning.

Ocean observing platforms (2023)

- Deployed ocean surface drifters and subsurface profiling floats in Indonesian seas.
- Analysed sea surface temperature, salinity profiles, currents and trajectories to evaluate platform lifespan and observe regional phenomena

Forecasting systems, data pipelines and dashboards (2025–2026)

- Helped define the solution for employing vortex-tracking algorithms and high-throughput data pipelines that stream real-time meteorological inputs into operational forecasting workflows.
- Supported the development of a web-based tropical cyclone forecasting and warning tool, in close collaboration with CLS (Collecte Localisation Satellites) France, letting forecasters refine tracks, adjust intensity parameters and publish advisories.
- Contributed to the front-end design and implementation of a maritime web dashboard integrating coastal and ship observations, numerical model output and satellite imagery.
- Managed hardware deployment across regional offices and training institutes, factory and site acceptance and user training.

Weather radar (C-band dual-polarization Doppler, Sidoarjo/Surabaya, 2024)

- Site preparation and civil works, including dismantling and lowering legacy radar and radome components.
- Installed and integrated primary power distribution, a 35 kVA diesel generator, a 20 kVA UPS, surge protection and grounding.
- Software configuration, signal calibration and issue resolution during Factory Acceptance Testing (FAT) review and final SAT.

Impact-based forecasting and precision agriculture pilot (2026)

- Developed protocols translating meteorological predictions into risk warnings for agriculture, water resources and disaster mitigation, working closely with CLS France.
- Led field operations with academic and research partners: soil-moisture sensor networks, drone-based spectral monitoring and crop digital-twin modelling.

Operations and business development (2022–2026)

- Analyse radar data and marine weather indicators to produce daily marine weather forecasts for commercial clients; install, configure and maintain marine meteorological systems for continuous data transmission.
- Completed manufacturer weather-radar training for meteorologists
- Supported business development, working closely with CLS France, by analysing marine-sector data to identify infrastructure expansion opportunities.

SKILLS

Technical & Field Engineering Skills

Meteorological Instrumentation | Equipment Installation | Sensor Calibration | System Integration

Data & Computational Skills

Python (computer programming) | Metocean Data Analysis

Subject-Matter Knowledge

Oceanography | Marine Meteorology | Climatology

Transversal & Operations Skills

Project Coordination | liaise with industrial professionals | liaise with engineers

LANGUAGE SKILLS

Mother tongue(s): **INDONESIAN**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C2	C1	B2	C1	B2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

PUBLICATIONS

2026

Rare interactions between marine heatwaves and tropical cyclones intensification in the Southern Indian Ocean

This study investigates the rare coupling of marine heatwaves and tropical cyclones in the southeastern Indian Ocean from 1982 to 2024, finding that only 8 out of 231 tropical cyclones interacted with active marine heatwaves. Despite their rarity, these heatwave-associated cyclones experienced significantly stronger intensification due to elevated ocean heat content, deeper warm water layers, and enhanced air–sea energy transfer.

2026

Impact of marine heatwaves and cold spells on coral reef ecosystem in a tropical region: a case study of Lombok Waters, Indonesia.

This paper analyzes thirty years of data (1993-2023) to show how climate phenomena like ENSO and the Indian Ocean Dipole cause significant sea level fluctuations in the Java Sea, decreasing levels by up to 1.4 meters during El Niño and

raising them by up to 2.1 meters during La Niña. These findings highlight the critical need for targeted coastal management strategies to address climate variability and urban vulnerability in Indonesia.

2025

Impact of marine heatwaves and cold spells on coral reef ecosystem in a tropical region: a case study of Lombok Waters, Indonesia

This study analyzes daily sea surface temperature data from 1998 to 2022 alongside satellite imagery and field surveys to examine how marine heatwaves and cold spells trigger severe coral bleaching in Indonesia's Lombok Waters. It demonstrates that extreme temperature anomalies led to a reduction in coral reef cover of roughly 30%, highlighting the urgent need for targeted management strategies to protect vulnerable marine ecosystems.

2024

Interannual variability of sea level anomaly (SLA) in the western Sumatra coastal waters driven by ENSO and IOD modulations

This paper investigates 31 years of data (1993-2023) to show how climate modes like ENSO and the Indian Ocean Dipole (IOD) drive sea level anomaly (SLA) variations along Sumatra's western coast. It reveals that SLA decreases during El Niño and positive IOD phases, while increasing during La Niña and negative IOD phases, with a one-month delayed response to ENSO and an immediate response to IOD.

2024

Effect of Marine Heat Waves for Coral Bleaching in Lombok Waters

This paper quantifies coral bleaching intensity in Lombok waters by analyzing satellite-derived sea surface temperatures and marine heatwaves over recent decades. It highlights dramatic fluctuations in local coral reef coverage, including a sharp collapse from over 36,000 hectares in 1998 down to under 4,900 hectares in 1999, driven by thermal stress alongside human activities and climate change.

2023

Marine Heatwaves in the Indonesian Fisheries Management Areas

This paper analyzes Daily Optimum Interpolation Sea Surface Temperature data from 1982 to 2020 to examine the characteristics and trends of Marine Heatwaves (MHWs) across 11 Indonesian Fisheries Management Areas. It reveals an overall increase in MHW frequency, duration, and intensity across these regions (particularly in areas like FMA 573, FMA 716, and FMA 711). This highlights the influence of climate drivers like ENSO and the Indian Ocean Dipole on local marine ecosystems.

2021

Sea surface temperature variability in Indonesia and its relation to regional climate indices

This paper analyzes monthly sea surface temperature data in Indonesian seas from 1979 to 2021 to examine its spatiotemporal patterns and relationships with climate indices. It finds that annual and semi-annual oscillations driven by the Asia-Australia monsoon system are the dominant SST drivers, with localized variability off western Sumatra tied to the Dipole Mode Index.